Annals of Hematology (2024) 103:593–602
https://doi.org/10.1007/s00277-023-05526-6

ORIGINAL ARTICLE

Non-viral pathogens of infectious diarrhoea post-allogeneic stem cell transplantation are associated with graft-versus-host disease

Matthew J. Rees[1] · Alexandra Rivalland[1] · Sarah Tan[1] · Mingdi Xie[1] · Michelle K. Yong[2,3,4,5] · David Ritchie[1]

Received: 4 July 2023 / Accepted: 26 October 2023 / Published online: 6 November 2023
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2023

Abstract
Infectious diarrhoea is common post-allogeneic haematopoietic stem-cell transplantation (alloHSCT). While the epidemiology of *Clostridioides difficile* infection (CDI) post-alloHSCT has been described, the impact of other diarrhoeal pathogens is uncertain. We reviewed all alloHSCT between 2017 and 2022 at a single large transplant centre; 374 patients were identified and included. The 1-year incidence of infectious diarrhoea was 23%, divided into viral (13/374, 3%), CDI (65/374, 17%) and other bacterial infections (16/374, 4%). There was a significant association between infectious diarrhoea within 1 year post-transplant and the occurrence of severe acute lower gastrointestinal graft-versus-host disease (GVHD, OR = 4.64, 95% CI 2.57–8.38, $p < 0.001$) and inferior GVHD-free, relapse-free survival on analysis adjusted for age, donor type, stem cell source and T-cell depletion (aHR = 1.64, 95% CI = 1.18–2.27, $p = 0.003$). When the classes of infectious diarrhoea were compared to no infection, bacterial (OR = 6.38, 95% CI 1.90–21.40, $p = 0.003$), CDI (OR = 3.80, 95% CI 1.91–7.53, $p < 0.001$) and multiple infections (OR = 11.16, 95% CI 2.84–43.92, $p < 0.001$) were all independently associated with a higher risk of severe GI GVHD. Conversely, viral infections were not (OR = 2.98, 95% CI 0.57–15.43, $p = 0.20$). Non-viral infectious diarrhoea is significantly associated with the development of GVHD. Research to examine whether the prevention of infectious diarrhoea via infection control measures or modulation of the microbiome reduces the incidence of GVHD is needed.

Keywords Graft-versus-host disease · Risk factors · Infectious diarrhoea · Clostrioides difficile infection · Stem-cell transplantation · Retrospective review

Introduction

Allogeneic haematopoietic stem-cell transplantation (alloHSCT) is a crucial therapy for patients with high-risk haematopoietic malignancies. The engraftment of allogeneic haematopoietic stem cells enables the donor immune system to eliminate tumour cells through the graft-versus-leukaemia effect [1]. Despite its therapeutic advantage, the beneficial impact of alloreactivity, which eliminates malignant cells, may be misdirected against host organs, in particular the liver, gut and skin, to cause graft-versus-host disease (GVHD). GVHD is a major obstacle to transplant outcomes as a primary cause of non-relapse mortality and morbidity through immunosuppression and infection [2, 3]. Despite prophylactic immunosuppressive agents, acute GVHD affects 10–50% of allograft recipients [3, 4]. Elements contributing to the risk and severity of GVHD include the degree of donor HLA mismatch, donor and recipient age, the conditioning regimen intensity and the GVHD prophylactic regimen [3, 5, 6].

Diarrhoea post-transplantation is ubiquitous with myriad causes. Allograft recipients are predisposed to infectious diarrhoea secondary to immunosuppression, prolonged hospitalisations and courses of broad-spectrum antibiotics as well as chemotherapy-related disruption of the gastrointestinal (GI) mucosal barrier [7, 8]. *Clostridiodies difficile*

✉ Matthew J. Rees
Matthew.Rees.MD@gmail.com

1 Clinical Haematology, Peter MacCallum Cancer Centre and Royal Melbourne Hospital, 305 Grattan St, Melbourne, VIC 3000, Australia

2 Department of Infectious Diseases, Peter MacCallum Cancer Centre, Melbourne, Australia

3 Victorian Infectious Diseases Service, Melbourne Health, Melbourne, Australia

4 Sir Peter MacCallum, Department of Oncology, University of Melbourne, Melbourne, Australia

5 National Centre for Infections in Cancer, Peter MacCallum Cancer Centre, Melbourne, Melbourne, Australia

infection (CDI), previously known as Clostridium difficile, is recognised as one of the most common causes of healthcare-associated infectious diarrhoea [9, 10], and its occurrence post-transplantation has been associated with the development of acute lower GI GVHD [7, 11–14]. This association has driven speculation that early infectious insults may precipitate or worsen GVHD through local tissue inflammation or altering microbial diversity. The latter is corroborated by the disruptive effect of infection with C. *difficile* and other enteric microbes on the gut microbiome [15–18].

While the epidemiology of C. *difficile* infection (CDI) post-alloHSCT has been described [7, 11–13], the frequency and impact of other diarrhoeal pathogens on GVHD are uncertain. We aimed to examine the epidemiology of all forms of infectious diarrhoea post-alloHSCT and explore their association with GVHD and post-transplant outcomes.

Method

Study design

A retrospective review of all patients undergoing alloHSCT between 2017 and 2022 was at Peter MacCallum Cancer Centre and the Royal Melbourne Hospital. Patients who received > 1 alloHSCT or syngeneic allografts were excluded.

Data source and collection

Hospital-written and computerised medical records were reviewed to obtain the following data: patient demographics, baseline disease characteristics, conditioning therapy and GVHD prophylaxis regimens. Microbiological specimen records were reviewed from the commencement of conditioning to 12 months post-alloHSCT. Infrequent haematological conditions for transplantation were grouped as 'other' and included severe aplastic anaemia, blastic plasmacytoid dendritic cell neoplasm, multiple myeloma and congenital immunodeficiencies.

Infectious diarrhoea

Infectious diarrhoea was defined as diarrhoea associated with the identification of an enteric pathogen on stool sample testing. Stool tests performed within 14 days of a previous test were considered part of the same infectious episode. Infections were categorised into CDI, non-clostridial bacterial, protozoal and viral aetiologies. Multiple infections refer to patients who experienced more than one form (CDI, non-clostridial bacterial, protozoal or viral) of infectious diarrhoea in the first year post-transplantation. Standard clinical care included the collection of a stool sample for microbiological testing at the onset of any new diarrhoeal episode post-transplant. All patients with first-onset new diarrhoea had faecal specimens cultured for C. *difficile*, Salmonella, Shigella and Campylobacter, as well as nucleic acid detection testing for norovirus. Patients who were culture positive for C. *difficile* underwent nucleic acid detection for C. *difficile* toxins A and B. CDI was only diagnosed in the presence of C. *difficile* toxin and diarrhoeal symptoms. From January 2018 onwards, the Biofire® multiplex PCR assay which detects nucleic acid of 19 pathogens associated with gastroenteritis (10 bacterial, four parasitic and five viral) was introduced; this test was performed on the first occasion for new community-onset diarrhoea in alloHSCT recipients.

Prophylactic fluoroquinolone antibiotic administration was not standard of care for the study patient population. Patients routinely received *Pneumocystis jirovecii* prophylaxis with either trimethoprim-sulfamethoxazole, dapsone or atovaquone. Broad spectrum antibiotics were administered during the peri-transplant period for febrile neutropenia, unexpected clinical deterioration or according to physician discretion. T-cell depletion was performed in vivo in all cases, as is standard practice at our institution. Prednisolone was occasionally added to ciclosporin and methotrexate as GVHD prophylaxis according to the following protocol [19].

End-points and statistical analysis

GVHD was graded according to Glucksberg and NIH criteria [20, 21]. The incidence of infectious diarrhoea in the first year post-alloHSCT was calculated as a cumulative incidence with death as a competing risk. Overall survival (OS) was defined as the date of transplantation to the date of death from any cause, censored on the date of the last follow-up. GVHD-free, relapse-free survival (GRFS) was defined as the date of transplantation to the date of grade III–IV acute GVHD, chronic GVHD requiring systemic immunosuppression, relapse or death and censored on the date of last follow-up.

Data were analysed using 'R' statistical software, version 4.2.2. Continuous and quantitative discrete variables were described as a median and interquartile range (IQR); categorical variables were described as a number and percentage. A two-sided p-value < 0.05 was considered statistically significant. Group comparisons were performed using the Chi-squared test. Variables related to the occurrence of GVHD and infectious diarrhoea were analysed using logistic regression. Univariate and multivariable analyses of variables associated with OS and GRFS were performed using the Cox proportional hazards regression model. A priori pre-specified prognostic factors included in the adjusted model were age, conditioning intensity, donor type (unrelated, sibling, haplo), stem cell source (peripheral blood, bone marrow, cord) and T-cell depletion of the allograft. Infectious diarrhoea was analysed as a time-dependent variable in all cox-modelling.

Results

Three hundred and seventy-four patients were identified and included in the study. Patient demographics, disease characteristics, GVHD prophylaxis and the rates of GVHD are summarised in Table 1. The median duration of follow-up was 31 months, and the 30-day mortality rate was 4.5% (17/374). The median time to onset of acute GVHD was 87 days (IQR 50, 132), excluding cases of late-onset acute GVHD; the median time to onset of acute classical GVHD was 57 days (IQR 38, 79). The median time to the onset of acute lower GI GVHD was 94 days (IQR 50, 133).

The incidence of infectious diarrhoea from the commencement of allogeneic conditioning to 1 year post-alloHSCT is shown in Table 2. At 1 year post-transplantation, 23% (86/374) of patients had experienced at least one episode of infectious diarrhoea. C. *difficile* was the most common pathogen to cause infectious diarrhoea, with 17% of patients (65/374) having at least one episode. The median time to CDI was 21 days (IQR 4, 70), and 25% of patients (16/65) experienced recurrent CDI. The 1-year incidence of infectious diarrhoea was not significantly different ($p=0.75$) for patients who underwent alloHSCT before and after the introduction of the Biofire® multiplex PCR assay, 25% (10/40) and 23% (76/334) respectively.

Four percent of patients (16/374) experienced a non-clostridial bacterial cause of diarrhoea, and 3% (13/374) experienced a viral cause. The median time to bacterial and viral diarrhoeal infections was 33 (IQR 3, 132) and 138 (IQR 83, 164) days, respectively (Fig. 1). There was only one case of a protozoal infection (Giardia Lamblia) at 161 days post-transplant. There were four cases of infectious diarrhoea caused by multiple pathogens simultaneously. Three of the cases included CDI, two norovirus and two Campylobacter species. One case subsequently developed acute lower GI GVHD 6 days later and one case occurred in the context of treatment for severe acute lower GI GVHD. There was no association between the occurrence of infectious diarrhoea or CDI and the age of the recipient ($p=0.63$, $p=0.69$ respectively), the intensity of the conditioning regimen ($p=0.89$, $p=0.85$), receipt of total body irradiation (TBI, $p=0.86$, $p=0.33$) or the underlying haematological condition ($p=0.71$, $p=0.12$).

The median duration of infectious diarrhoea was 5 days (range 1, 26 days). There was no association between the duration of infectious diarrhoea and the development of severe acute lower GI GVHD ($p=0.63$) or chronic GI GVHD ($p=0.23$). Of patients who developed infectious diarrhoea, 62% of CDI, 47% of bacterial and 33% of viral infections had received broad-spectrum antibiotics in the preceding month. In patients who experienced infectious diarrhoea, there was no difference in the rate of severe acute lower GI GVHD, 45 vs 55%, $p=0.15$, for patients who did or did not receive broad spectrum antibiotics.

There was a significant relationship between the occurrence of infectious diarrhoea within 1 year post-transplant and the occurrence of severe acute lower GI GVHD (OR = 4.64, 95% CI 2.57–8.38, $p<0.001$) and chronic GI GVHD (OR = 2.82, 95% CI 1.37–5.78, $p=0.003$) (Fig. 2). The majority of patients (93%, 54/58) diagnosed with severe acute lower GI GVHD underwent a biopsy; of these patient, the biopsy confirmed a diagnosis of GVHD in 81% (44/54), was non-diagnostic in 9% (5/54) and was negative in 9% (5/54) of cases.

The 1-year probability of developing grade 3 or higher acute lower GI GVHD was 37% in patients with an episode of infectious diarrhoea and 12% in those without. Twenty-nine patients experienced both acute severe lower GI GVHD and an episode of infectious diarrhoea; in 79% (23/29) of cases, the episode of infectious diarrhoea preceded or occurred simultaneous to the development of GI GVHD. In this group of patients, the median time to onset severe lower GI GVHD post-infectious diarrhoea was 38 days (IQR 7, 79). When the classes of infectious diarrhoea were compared to no infection, bacterial (OR = 6.38, 95% CI 1.90–21.40, $p=0.003$), CDI (OR = 3.80, 95% CI 1.91–7.53, $p<0.001$) and multiple infections (OR = 11.16, 95% CI 2.84–43.92, $p<0.001$) were all independently associated with a higher risk of severe lower GI GVHD. Conversely, viral infections were not (OR = 2.98, 95% CI 0.57–15.43, $p=0.20$). There was a significant association between patients who experienced infectious diarrhoea within 1 year post-transplant and any form of acute GVHD (OR = 2.34, 95% CI 1.42–3.85, $p<0.001$).

Of 86 patients who developed infectious diarrhoea, 82 (95%) were receiving a calcineurin inhibitor (CNI) at the time: tacrolimus (TAC) in 10 (12%) cases and ciclosporin (CSA) in 72 (84%) cases. In 36 patients receiving intravenous CSA, the median trough level was 252 ng/mL (range 107, 720); 6 patients (16%) had a CSA trough levels < 175 ng/mL, and 4 (11%) had CSA trough levels between 175 and 199 ng/mL. In 36 patients receiving oral CSA, the median 2-h peak level was 424 ng/mL (range 131, 1832); 20 patients (56%) had CSA peak levels < 800 ng/mL, and 6 (16%) had CSA peak levels between 800 and 1199 ng/mL. In 3 patients receiving intravenous tacrolimus, the median trough level was 6.6 ng/mL (range 6, 9.1) and in 7 patients receiving oral tacrolimus, the median trough level was 7.6 ng/mL (range 4.7, 22.5). Only 1 patient receiving tacrolimus had a suboptimal drug level (4.7 ng/mL). The rate of acute lower GI GVHD was higher in patients with

Table 1 Patient demographics, disease characteristics, conditioning regimen, graft-versus-host disease prophylaxis and the incidence of graft-versus-host disease according to patients who experienced an episode of infectious diarrhoea within 1 year post-transplantation

Variable	Infectious diarrhoea $n=86$	No infectious diarrhoea $n=288$	All $n=374$
Age, years	54 (18, 75)	53 (16, 74)	53 (16, 75)
Sex, male	56 (65%)	181 (63%)	237 (63%)
Haematological condition			
- Acute leukaemia	48 (56%)	138 (48%)	186 (50%)
- MDS/CMML	9 (10%)	44 (15%)	53 (14%)
- Lymphoma	17 (20%)	61 (21%)	78 (21%)
- Myeloproliferative neoplasm	8 (9%)	30 (10%)	38 (10%)
- Other	4 (5%)	15 (5%)	19 (5%)
Conditioning intensity			
- Myeloablative	31 (36%)	98 (34%)	129 (34%)
- Non-myeloablative	4 (5%)	11 (4%)	15 (4%)
- Reduced intensity	51 (59%)	179 (62%)	230 (62%)
Total body irradiation			
- None	66 (77%)	221 (77%)	287 (77%)
- Low-dose (≤ 6 Gy)	7 (8%)	28 (10%)	35 (9%)
- High-dose (≥ 12 Gy)	13 (15%)	39 (13%)	52 (14%)
Allograft source			
- Haploidentical	38 (44%)	46 (16%)	58 (16%)
- Sibling	12 (14%)	90 (31%)	128 (34%)
- Unrelated	36 (42%)	152 (53%)	188 (50%)
Graft type			
- Bone marrow	5 (6%)	13 (4.5%)	18 (5%)
- Peripheral blood	80 (93%)	276 (96%)	355 (95%)
- Cord blood	1 (1%)	0	1 (0%)
GVHD prophylaxis			
- T-cell depletion	30 (35%)	136 (47%)	166 (44%)
- Prednisolone	19 (22%)	48 (16%)	67 (18%)
- Methotrexate	0	1 (0%)	1 (0%)
- Methotrexate + mycophenolate	0	1 (0%)	1 (0%)
- TAC + mycophenolate	9 (10%)	33 (11%)	42 (11%)
- CSA	4 (5%)	4 (1%)	8 (2%)
- CSA + methotrexate	69 (80%)	238 (83%)	307 (82%)
- CSA + mycophenolate	4 (5%)	11 (4%)	15 (4%)
Acute GVHD[a]			
- None	46 (53%)	210 (73%)	256 (68%)
- Grades 1–2	6 (7%)	44 (15%)	50 (13%)
- Grades 3–4	34 (40%)	34 (12%)	68 (18%)
Acute lower GI GVHD staging[a]			
- None	52 (60%)	248 (86%)	300 (80%)
- Grades 1–2	5 (6%)	11 (4%)	16 (4%)
- Grades 3–4	29 (34%)	29 (10%)	58 (16%)
Chronic GVHD grade[b]			
- None	48 (56%)	186 (65%)	234 (62%)
- Mild	16 (19%)	47 (16%)	63 (17%)
- Moderate	9 (10%)	29 (10%)	38 (10%)
- Severe	13 (15%)	26 (9%)	39 (10%)
Chronic GI GVHD	16 (19%)	22 (8%)	38 (10%)

Unless otherwise specified, variables are reported as median (range)

'Other' includes blastic plasmacytoid dendritic cell neoplasms, multiple myeloma and congenital immunodeficiencies

ALL acute lymphoblastic leukaemia, *AML* acute myeloid leukaemia, *MDS* myelodysplastic syndrome, *CMML* chronic myelomonocytic leukaemia, *CSA* ciclosporin, *TAC* tacrolimus, *GI* gastrointestinal, *Gy* Gray units

[a]Graded according to Glucksberg criteria

[b]According to NIH criteria

Table 2 The 1-year incidence and causative pathogens of infectious diarrhoea in 374 allogeneic haematopoietic stem cell transplant recipients

Variable	
Number of infections within 1 year post-transplant	
- None	288 (77%)
- 1 infection	60 (16%)
- 2 infections	21 (6%)
- 3 infections	3 (1%)
- ≥4 infections	2 (1%)
Microbiological cause of primary episode of infectious diarrhoea ($n = 86$)	
- *Adenovirus*	1 (1%)
- *Aeromonas* species	3 (3%)
- *Campylobacter* species	6 (7%)
- *Clostridioides difficile*	62 (72%)
- *Escherichia coli* species	3 (3%)
- *Norovirus*	8 (9%)
- *Salmonella* species	1 (1%)
- *Yersinia enterocolitica*	2 (2%)
Time to the first episode of infectious diarrhoea, days	22 (−5, 362)

Unless otherwise specified, variables are reported as median (range)

suboptimal or low CNI drug levels at the time of infectious diarrhoea compared to those with therapeutic levels, 37% vs 22%; however, this was not statistically significant ($p = 0.68$).

Experiencing an episode of infectious diarrhoea within 1 year of transplantation was significantly associated with inferior GRFS on unadjusted cox-proportional hazards modelling (HR = 1.60, 95% CI = 1.16–2.21, $p = 0.004$) and on analysis adjusted for age, donor type, stem cell source and T-cell depletion (aHR = 1.64, 95% CI = 1.18–2.27, $p = 0.003$) (Fig. 3). There was no significant association between experiencing an episode of infectious diarrhoea within 1 year of transplant and overall survival on unadjusted (OS, HR = 1.30, 95% CI = 0.85–1.99, $p = 0.22$) and adjusted analysis (aHR = 1.41, 95% CI = 0.92–2.16, $p = 0.12$).

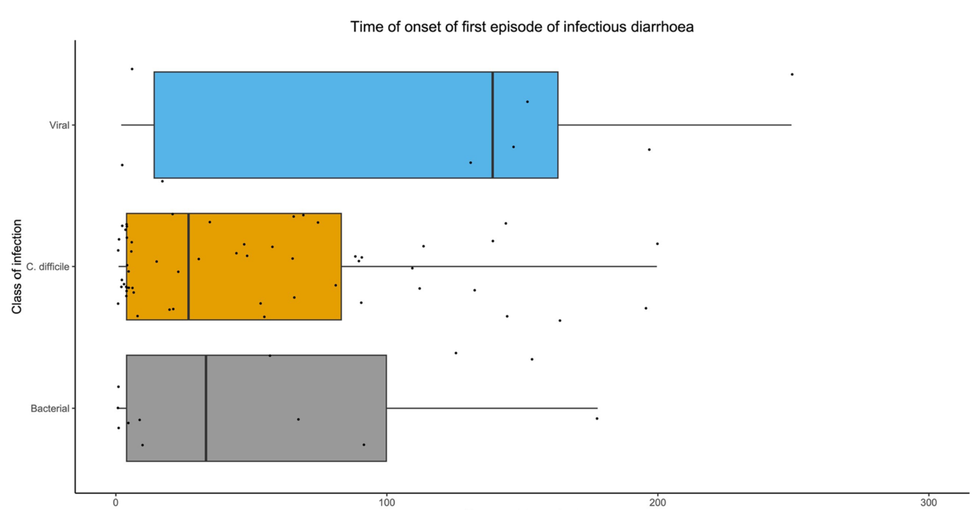
Discussion

We provide a comprehensive assessment of the epidemiology and morbidity of infectious diarrhoea post-alloHSCT, with nearly one in four recipients experiencing an infection in the first-year post-transplant [7, 22], and such patients experiencing inferior GRFS and being over four times more likely to develop severe lower GI GVHD. Our study is the first to show that GVHD risk varies according to the diarrhoeal pathogen responsible.

While rarely fatal in isolation, infectious diarrhoea affects one-quarter to one-third of patients' early post-alloHSCT and is associated with increased non-relapse mortality [8, 22]. Since the 1980s, C. *difficile* has prevailed as the principle cause of infectious diarrhoea post-transplant, affecting 12–27% of patients [7, 8, 11, 12, 22]. CDI primarily occurs in the pre- and early post-engraftment periods [8, 12, 14], a time when patients experience neutropenia, antimicrobial exposure and chemotherapy-related disruptions of the enteric mucosal barrier. Recalcitrant and protracted CDI are common in the transplant population who often receive antibiotics and corticosteroids [23, 24], with one-quarter of patients in our cohort experiencing a recurrence. Risk factors for CDI post-alloHSCT have been inconsistently reported with poor reproducibility but include the type of conditioning used, the number of prior lines of chemotherapy and the receipt of broad-spectrum antibiotics [7, 8, 12–14, 25]. In this study, we did not identify an association between CDI and the conditioning regimen or the number of prior lines of chemotherapy.

Besides CDI, bacteria and viruses are responsible for a significant proportion of infectious diarrhoea post-transplant,

Fig. 1 Boxplot of onset of infectious diarrhoea according to pathogen type (C. *difficile*, viral, non-clostridial bacterial). Black dots represent individual data points



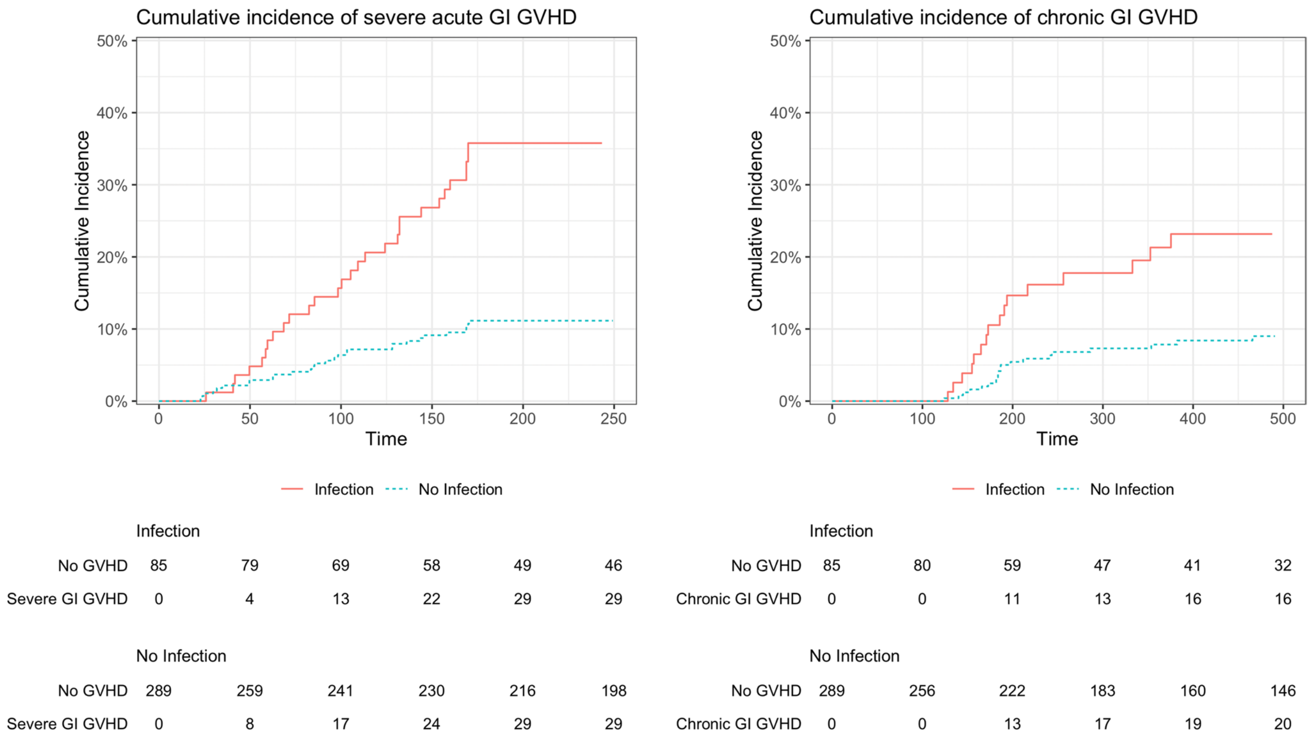
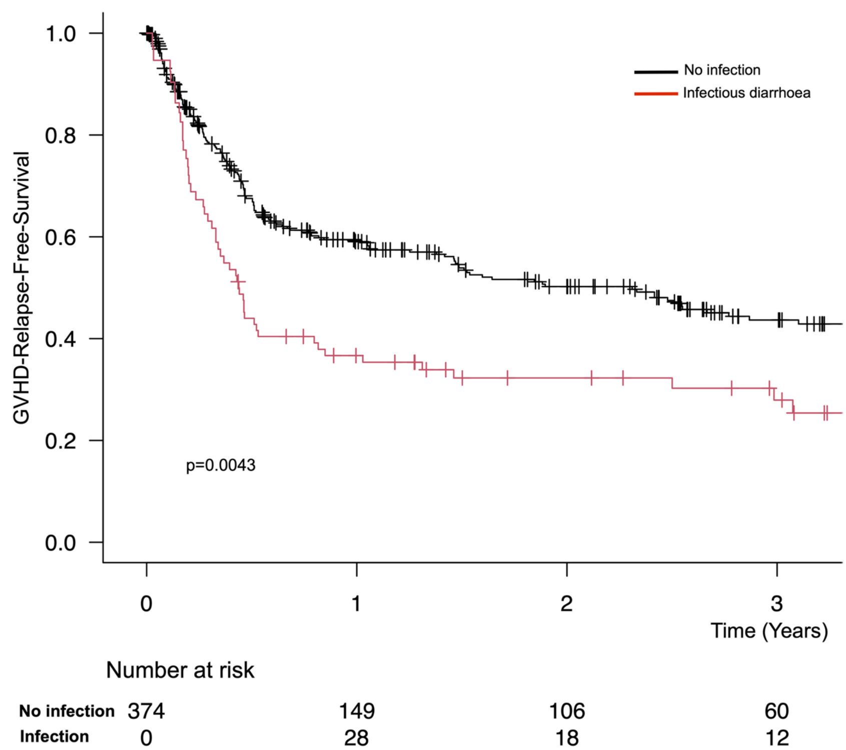


Fig. 2 The cumulative incidence of acute lower and chronic graft-versus-host disease of the gastrointestinal tract for patients who did and did not experience an episode of infectious diarrhoea within the first year post-transplant

Fig. 3 Simon and Makuch plot of GVHD-free, relapse-free survival stratified according to patients who did and did not experience an episode of infectious diarrhoea within 1 year post-transplant



accounting for a total of 8% of infections. While high, this is significantly less than reported in other studies using sensitive multiplex PCR-based assays in which bacterial and viral pathogens each have a 1 year post-alloHSCT cumulative incidence of ~ 15% [8, 26]. While such assays can significantly augment the identification of infectious aetiologies, the pathogenic role of some pathogens, such as enteropathogenic *Escherichia coli*, remains to be confirmed [26]. Compared to CDI, non-clostridial bacterial and viral causes typically occur late post-engraftment, potentially due to increased exposure to these pathogens in the ambulatory setting [8].

We demonstrate a strong association between early infectious diarrhoea, inferior GRFS and acute and chronic GVHD. In line with previous reports, the GI tract was the most common site of GVHD associated with infectious diarrhoea [7, 8, 12, 13]. Infectious diarrhoea preceded a diagnosis of GI GVHD in most cases, supporting a formative role in GVHD pathogenesis [14]. The function of damage and pathogen-associated molecular patterns (PAMPs), as well as a dysfunctional microbiome in the development of GVHD, provides a plausible biological explanation for this link [27–29]. Tissue damage and PAMP activation lead to the release of pro-inflammatory cytokines, which in turn promote donor T cell activation, differentiation, proliferation and migration to the gastrointestinal tract, ultimately leading to T-cell recognition of host-antigens [30]. Notably, the association between CDI and GVHD may alternatively be explained by CDI being a surrogate for gut dysbiosis, which is a known risk factor for GVHD [31].

While not statistically significant, patients with low oral CNI drug levels experienced a numerically higher rate of acute lower GI GVHD. Importantly, 72% of patients receiving oral CSA at the time of infectious diarrhoea had suboptimal therapeutic drug levels compared to just 1 patient receiving oral TAC. This is likely a consequence of differences in drug bioavailability, the GI absorption of TAC does not depend on the presence of bile salts and its bioavailability increases with mild to moderate diarrhoea [32].

Fifty years after the association between the gut microbiome and GVHD was first recognised [33], we now understand how the composition of the microbiome influences post-transplant outcomes and how it can be modulated therapeutically by faecal microbiota transplantation (FMT) [34–37]. The interplay between the intestinal microbiota and the mucosal immune system is central to preventing infections [38, 39], maintaining mucosal integrity and producing immune-regulatory metabolites [40, 41]. Increased bacterial diversity post-alloHSCT has been associated with reduced GVHD; specifically, an abundance of anaerobic commensals, including bacteria from the genus *Blautia*, is associated with less frequent and severe GVHD [34, 42]. AlloHSCT damages the microbiome through antibiotic and antacid exposure, intestinal inflammation from chemoradiotherapy and changes in diet [43]. Infection by C. *difficile* and other enteric pathogens also detrimentally impacts microbial composition and function [15–18]. Distinct from previous research, we confirm an association between non-clostridial bacterial forms of diarrhoea and GVHD but not viral forms of diarrhoea. The explanation for this is unclear, but it may be a consequence of the delayed onset of viral infections.

The management of infectious diarrhoea post-transplant encompasses infection control, supportive care measures, and, where appropriate, steps to preserve or restore gut microbial diversity. FMT, which refers to administering faecal matter from a donor to a recipient with the intent of modifying the composition of the recipient's microbiome, is an emerging therapy to restore microbiome diversity in alloHSCT recipients. FMT has been introduced for allograft recipients as a therapeutic intervention for GVHD and recurrent CDI, as well as to prevent the dysbiosis inherent to transplantation and thereby reduce microbiome-related complications such as CDI, GVHD or gut colonisation with antimicrobial-resistant bacteria [37, 44–47].

The role of FMT as a treatment for recurrent CDI is established, and reports of its use in alloHSCT recipients suggest that it is safe and effective [47, 48]. Encouraging data exists for FMT as a treatment for steroid-resistant gut GVHD with overall response rates of 43–75%. However, this needs to be confirmed in larger prospective studies [46, 49] and practical issues concerning the optimal FMT delivery method, use of antibiotics and donor choice remain to be decided. FMT entails risks from procedural complications and the hazard of transmitting infection to an immunocompromised host with impaired gut wall integrity. While data endorses FMT as safe, the concern for infection means efforts to screen donor stools to exclude potentially transmittable pathogens is crucial [45].

This research is subject to all the limitations inherent to a retrospective study. As a chart review, we could not confirm that all diarrhoeal episodes met standard criteria (i.e. ≥ three unformed stools/day); thus, our definition was based on submitting a stool sample for diagnostic testing. We acknowledge the significant possibility of testing bias introduced by patients undergoing microbiological testing while being investigated for GI GVHD and the potential for false positive results given the high rates of diarrhoea and testing in the peri-transplant period. Another relevant limitation is the conventional microbiological stool assays used for 1 year of the study period, which have less sensitivity for detecting many diarrhoeal pathogens [26].

In summary, this single-centre experience illustrates that infectious diarrhoea is a frequent, early complication post-transplant which is significantly associated with the development of GVHD and inferior GRFS. Research is still needed to examine whether the prevention of infectious diarrhoea may reduce the incidence of GVHD; however, in this vulnerable population, ensuring optimal compliance with infection prevention techniques, such as hand hygiene, isolation precautions and adequate environmental cleaning, is imperative. Investigation of innovative approaches, such as FMT, which modulate the gut microbiome to treat and prevent infectious diarrhoea and GVHD, are ongoing but hold promise to improve patient outcomes.

Author contribution Matthew James Rees, Michelle Yong and David Ritchie designed the study; All authors contributed data and reviewed the manuscript; Matthew James Rees analysed the data; Matthew James Rees wrote the first draft of the manuscript, and all authors read and approved the final manuscript.

Data availability The dataset generated and analysed during the current study is available from the corresponding author upon reasonable request.

Declarations

Ethics approval This retrospective study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the ethics committees of Peter MacCallum Cancer Centre and the Royal Melbourne hospital.

Competing interests The authors declare no competing interests.

Disclosures The authors have no disclosures.

References

1. Kolb HJ, Mittermüller J, Clemm C, Holler E, Ledderose G, Brehm G, Heim M, Wilmanns W (1990) Donor leukocyte transfusions for treatment of recurrent chronic myelogenous leukemia in marrow transplant patients. Blood 76(12):2462–2465
2. Jagasia M, Arora M, Flowers MED, Chao NJ, McCarthy PL, Cutler CS, Urbano-Ispizua A, Pavletic SZ, Haagenson MD, Zhang M-J, Antin JH, Bolwell BJ, Bredeson C, Cahn J-Y, Cairo M, Gale RP, Gupta V, Lee SJ, Litzow M, Weisdorf DJ, Horowitz MM, Hahn T (2012) Risk factors for acute GVHD and survival after hematopoietic cell transplantation. Blood 119(1):296–307. https://doi.org/10.1182/blood-2011-06-364265
3. El-Jawahri A, Li S, Antin JH, Spitzer TR, Armand PA, Koreth J, Nikiforow S, Ballen KK, Ho VT, Alyea EP, Dey BR, McAfee SL, Glotzbecker BE, Soiffer RJ, Cutler CS, Chen Y-B (2016) Improved treatment-related mortality and overall survival of patients with grade IV acute GVHD in the modern years. Biol Blood Marrow Transplant 22(5):910–918. https://doi.org/10.1016/j.bbmt.2015.12.024
4. Weisdorf D, Zhang M-J, Arora M, Horowitz MM, Rizzo JD, Eapen M (2012) Graft-versus-host disease induced graft-versus-leukemia effect: greater impact on relapse and disease-free survival after reduced intensity conditioning. Biol Blood Marrow Transplant 18(11):1727–1733. https://doi.org/10.1016/j.bbmt.2012.06.014
5. Garnett C, Apperley JF, Pavlů J (2013) Treatment and management of graft-versus-host disease: improving response and survival. Ther Adv Hematol 4(6):366–378. https://doi.org/10.1177/2040620713489842
6. Hahn T, McCarthy PL Jr, Zhang MJ, Wang D, Arora M, Frangoul H, Gale RP, Hale GA, Horan J, Isola L, Maziarz RT, van Rood JJ, Gupta V, Halter J, Reddy V, Tiberghien P, Litzow M, Anasetti C, Pavletic S, Ringdén O (2008) Risk factors for acute graft-versus-host disease after human leukocyte antigen-identical sibling transplants for adults with leukemia. J Clin Oncol 26(35):5728–5734. https://doi.org/10.1200/jco.2008.17.6545
7. Alonso CD, Treadway SB, Hanna DB, Huff CA, Neofytos D, Carroll KC, Marr KA (2012) Epidemiology and outcomes of Clostridium difficile infections in hematopoietic stem cell transplant recipients. Clin Infect Dis 54(8):1053–1063. https://doi.org/10.1093/cid/cir1035
8. Van Praet JT, Huysman A, De Knijf E, De Buyser S, Snauwaert S, Van Droogenbroeck J, Lodewyck T, Schauwvlieghe A, Selleslag D, Reynders M (2023) Infectious diarrhea after allogeneic hematopoietic cell transplantation assessed by a multiplex polymerase chain reaction assay. Int J Infect Dis 127:17–22. https://doi.org/10.1016/j.ijid.2022.11.045
9. Kelly CP, LaMont JT (2008) Clostridium difficile–more difficult than ever. N Engl J Med 359(18):1932–1940. https://doi.org/10.1056/NEJMra0707500
10. Slimings C, Armstrong P, Beckingham WD, Bull AL, Hall L, Kennedy KJ, Marquess J, McCann R, Menzies A, Mitchell BG, Richards MJ, Smollen PC, Tracey L, Wilkinson IJ, Wilson FL, Worth LJ, Riley TV (2014) Increasing incidence of Clostridium difficile infection, Australia, 2011–2012. Med J Aust 200(5):272–276. https://doi.org/10.5694/mja13.11153
11. Chung S, Abou Mourad Y (2019) Clostridium difficile infection among hematopoietic stem cell transplant recipients and subsequent development of graft-versus-host disease. Blood 134:4494. https://doi.org/10.1182/blood-2019-129663
12. Bhutani D, Jaiyeoba C, Kim S, Naylor P, Uberti JP, Ratanatharathorn V, Ayash L, Deol A, Alavi A, Revankar S, Chandrasekar P (2019) Relationship between clostridium difficile infection and gastrointestinal graft versus host disease in recipients of allogeneic stem cell transplantation. Bone Marrow Transplant 54(1):164–167. https://doi.org/10.1038/s41409-018-0270-x
13. Chakrabarti S, Lees A, Jones SG, Milligan DW (2000) Clostridium difficile infection in allogeneic stem cell transplant recipients is associated with severe graft-versus-host disease and non-relapse mortality. Bone Marrow Transplant 26(8):871–876. https://doi.org/10.1038/sj.bmt.1702627
14. Ramanathan M, Kim S, He N, Chen M, Hematti P, Abid MB, Rotz SJ, Williams KM, Lazarus HM, Wirk B, Yin DE, Kanakry CG, Perales M-A, Chemaly RF, Dandoy CE, Riches M, Ustun C (2023) The incidence and impact of clostridioides difficile infection on transplant outcomes in acute leukemia and MDS after allogeneic hematopoietic cell transplant—a CIBMTR study. Bone Marrow Transplant 58(4):360–366. https://doi.org/10.1038/s41409-022-01896-z
15. De La Cochetière MF, Durand T, Lalande V, Petit JC, Potel G, Beaugerie L (2008) Effect of antibiotic therapy on human fecal microbiota and the relation to the development of Clostridium difficile. Microb Ecol 56(3):395–402. https://doi.org/10.1007/s00248-007-9356-5
16. Sehgal K, Khanna S (2021) Gut microbiome and Clostridioides difficile infection: a closer look at the microscopic interface. Therap Adv Gastroenterol 14:1756284821994736. https://doi.org/10.1177/1756284821994736
17. Rea MC, O'Sullivan O, Shanahan F, O'Toole PW, Stanton C, Ross RP, Hill C (2012) Clostridium difficile carriage in elderly subjects and associated changes in the intestinal microbiota. J Clin Microbiol 50(3):867–875. https://doi.org/10.1128/jcm.05176-11
18. Nelson AM, Walk ST, Taube S, Taniuchi M, Houpt ER, Wobus CE, Young VB (2012) Disruption of the human gut microbiota following Norovirus infection. PLoS ONE 7(10):e48224. https://doi.org/10.1371/journal.pone.0048224
19. Hoyt R, Ritchie DS, Roberts AW, Macgregor L, Curtis DJ, Szer J, Grigg AP (2008) Cyclosporin, methotrexate and prednisolone for graft-versus-host disease prophylaxis in allogeneic peripheral blood progenitor cell transplants. Bone Marrow Transplant 41(7):651–658. https://doi.org/10.1038/sj.bmt.1705955

20. Glucksberg H, Storb R, Fefer A, Buckner CD, Neiman PE, Clift RA, Lerner KG, Thomas ED (1974) Clinical manifestations of graft-versus-host disease in human recipients of marrow from HL-A-matched sibling donors. Transplantation 18(4):295–304. https://doi.org/10.1097/00007890-197410000-00001
21. Jagasia MH, Greinix HT, Arora M, Williams KM, Wolff D, Cowen EW, Palmer J, Weisdorf D, Treister NS, Cheng GS, Kerr H, Stratton P, Duarte RF, McDonald GB, Inamoto Y, Vigorito A, Arai S, Datiles MB, Jacobsohn D, Heller T, Kitko CL, Mitchell SA, Martin PJ, Shulman H, Wu RS, Cutler CS, Vogelsang GB, Lee SJ, Pavletic SZ, Flowers ME (2015) National Institutes of Health Consensus Development Project on criteria for clinical trials in chronic graft-versus-host disease: I. The (2014) Diagnosis and Staging Working Group report. Biol Blood Marrow Transplant 21(3):389-401.e381. https://doi.org/10.1016/j.bbmt.2014.12.001
22. Yolken RH, Bishop CA, Townsend TR, Bolyard EA, Bartlett J, Santos GW, Saral R (1982) Infectious gastroenteritis in bone-marrow-transplant recipients. N Engl J Med 306(17):1009–1012. https://doi.org/10.1056/nejm198204293061701
23. Kelly CP (2012) Can we identify patients at high risk of recurrent Clostridium difficile infection? Clin Microbiol Infect 18(Suppl 6):21–27. https://doi.org/10.1111/1469-0691.12046
24. Ma GK, Brensinger CM, Wu Q, Lewis JD (2017) Increasing incidence of multiply recurrent clostridium difficile infection in the United States: a cohort study. Ann Intern Med 167(3):152–158. https://doi.org/10.7326/m16-2733
25. Kinnebrew MA, Lee YJ, Jenq RR, Lipuma L, Littmann ER, Gobourne A, No D, van den Brink M, Pamer EG, Taur Y (2014) Early Clostridium difficile infection during allogeneic hematopoietic stem cell transplantation. PLoS ONE 9(3):e90158. https://doi.org/10.1371/journal.pone.0090158
26. Rogers WS, Westblade LF, Soave R, Jenkins SG, van Besien K, Singh HK, Walsh TJ, Small CB, Shore T, Crawford CV, Satlin MJ (2020) Impact of a multiplexed polymerase chain reaction panel on identifying diarrheal pathogens in hematopoietic cell transplant recipients. Clin Infect Dis 71(7):1693–1700. https://doi.org/10.1093/cid/ciz1068
27. Zeiser R (2019) Advances in understanding the pathogenesis of graft-versus-host disease. Br J Haematol 187(5):563–572. https://doi.org/10.1111/bjh.16190
28. Zeiser R, Blazar BR (2017) Acute graft-versus-host disease - biologic process, prevention, and therapy. N Engl J Med 377(22):2167–2179. https://doi.org/10.1056/NEJMra1609337
29. Klämbt V, Wohlfeil SA, Schwab L, Hülsdünker J, Ayata K, Apostolova P, Schmitt-Graeff A, Dierbach H, Prinz G, Follo M, Prinz M, Idzko M, Zeiser R (2015) A novel function for P2Y2 in myeloid recipient-derived cells during graft-versus-host disease. J Immunol 195(12):5795–5804. https://doi.org/10.4049/jimmunol.1501357
30. Choi SW, Levine JE, Ferrara JL (2010) Pathogenesis and management of graft-versus-host disease. Immunol Allergy Clin North Am 30(1):75–101. https://doi.org/10.1016/j.iac.2009.10.001
31. Fredricks DN (2019) The gut microbiota and graft-versus-host disease. J Clin Investig 129(5):1808–1817. https://doi.org/10.1172/JCI125797
32. Lemahieu W, Maes B, Verbeke K, Rutgeerts P, Geboes K, Vanrenterghem Y (2005) Cytochrome P450 3A4 and P-glycoprotein activity and assimilation of tacrolimus in transplant patients with persistent diarrhea. Am J Transplant 5(6):1383–1391. https://doi.org/10.1111/j.1600-6143.2005.00844.x
33. Bekkum DWV, Roodenburg J, Heidt PJ, Waaij DVD (1974) Mitigation of secondary disease of allogeneic mouse radiation chimeras by modification of the intestinal microflora. JNCI 52(2):401–404. https://doi.org/10.1093/jnci/52.2.401
34. Burgos da Silva M, Ponce DM, Dai A, M. Devlin S, Gomes ALC, Moore G, Slingerland J, Shouval R, Armijo GK, DeWolf S, Fei T, Clurman A, Fontana E, Amoretti LA, Wright RJ, Andrlova H, Miltiadous O, Perales M-A, Taur Y, Peled JU, van den Brink MRM (2022) Preservation of the fecal microbiome is associated with reduced severity of graft-versus-host disease. Blood 140(22):2385–2397. https://doi.org/10.1182/blood.2021015352
35. Taur Y, Jenq RR, Perales M-A, Littmann ER, Morjaria S, Ling L, No D, Gobourne A, Viale A, Dahi PB, Ponce DM, Barker JN, Giralt S, van den Brink M, Pamer EG (2014) The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. Blood 124(7):1174–1182. https://doi.org/10.1182/blood-2014-02-554725
36. Payen M, Nicolis I, Robin M, Michonneau D, Delannoye J, Mayeur C, Kapel N, Berçot B, Butel M-J, Le Goff J, Socié G, Rousseau C (2020) Functional and phylogenetic alterations in gut microbiome are linked to graft-versus-host disease severity. Blood Adv 4(9):1824–1832. https://doi.org/10.1182/bloodadvances.2020001531
37. DeFilipp Z, Hohmann E, Jenq RR, Chen YB (2019) Fecal microbiota transplantation: restoring the injured microbiome after allogeneic hematopoietic cell transplantation. Biol Blood Marrow Transplant 25(1):e17–e22. https://doi.org/10.1016/j.bbmt.2018.10.022
38. Vollaard EJ, Clasener HA (1994) Colonization resistance. Antimicrob Agents Chemother 38(3):409–414. https://doi.org/10.1128/aac.38.3.409
39. Kamada N, Chen GY, Inohara N, Núñez G (2013) Control of pathogens and pathobionts by the gut microbiota. Nat Immunol 14(7):685–690. https://doi.org/10.1038/ni.2608
40. Smith PM, Howitt MR, Panikov N, Michaud M, Gallini CA, Bohlooly YM, Glickman JN, Garrett WS (2013) The microbial metabolites, short-chain fatty acids, regulate colonic Treg cell homeostasis. Science 341(6145):569–573. https://doi.org/10.1126/science.1241165
41. Donohoe DR, Garge N, Zhang X, Sun W, O'Connell TM, Bunger MK, Bultman SJ (2011) The microbiome and butyrate regulate energy metabolism and autophagy in the mammalian colon. Cell Metab 13(5):517–526. https://doi.org/10.1016/j.cmet.2011.02.018
42. Jenq RR, Taur Y, Devlin SM, Ponce DM, Goldberg JD, Ahr KF, Littmann ER, Ling L, Gobourne AC, Miller LC, Docampo MD, Peled JU, Arpaia N, Cross JR, Peets TK, Lumish MA, Shono Y, Dudakov JA, Poeck H, Hanash AM, Barker JN, Perales M-A, Giralt SA, Pamer EG, van den Brink MRM (2015) Intestinal blautia is associated with reduced death from graft-versus-host disease. Biol Blood Marrow Transplant 21(8):1373–1383. https://doi.org/10.1016/j.bbmt.2015.04.016
43. Shono Y, van den Brink MRM (2018) Gut microbiota injury in allogeneic haematopoietic stem cell transplantation. Nat Rev Cancer 18(5):283–295. https://doi.org/10.1038/nrc.2018.10
44. Bilinski J, Grzesiowski P, Sorensen N, Madry K, Muszynski J, Robak K, Wroblewska M, Dzieciatkowski T, Dulny G, Dwilewicz-Trojaczek J, Wiktor-Jedrzejczak W, Basak GW (2017) Fecal microbiota transplantation in patients with blood disorders inhibits gut colonization with antibiotic-resistant bacteria: results of a prospective, single-center study. Clin Infect Dis 65(3):364–370. https://doi.org/10.1093/cid/cix252
45. Pession A, Zama D, Muratore E, Leardini D, Gori D, Guaraldi F, Prete A, Turroni S, Brigidi P, Masetti R (2021) Fecal microbiota transplantation in allogeneic hematopoietic stem cell transplantation recipients: a systematic review. J Pers Med 11 (2). https://doi.org/10.3390/jpm11020100
46. van Lier YF, Davids M, Haverkate NJE, de Groot PF, Donker ML, Meijer E, Heubel-Moenen FCJI, Nur E, Zeerleder SS, Nieuwdorp M, Blom B, Hazenberg MD (2020) Donor fecal microbiota transplantation ameliorates intestinal graft-versus-host disease in allogeneic hematopoietic cell transplant recipients. Sci Transl Med 12 (556). https://doi.org/10.1126/scitranslmed.aaz8926

47. Webb BJ, Brunner A, Ford CD, Gazdik MA, Petersen FB, Hoda D (2016) Fecal microbiota transplantation for recurrent Clostridium difficile infection in hematopoietic stem cell transplant recipients. Transpl Infect Dis 18(4):628–633. https://doi.org/10.1111/tid.12550
48. Cammarota G, Ianiro G, Tilg H, Rajilić-Stojanović M, Kump P, Satokari R, Sokol H, Arkkila P, Pintus C, Hart A, Segal J, Aloi M, Masucci L, Molinaro A, Scaldaferri F, Gasbarrini G, Lopez-Sanroman A, Link A, de Groot P, de Vos WM, Högenauer C, Malfertheiner P, Mattila E, Milosavljević T, Nieuwdorp M, Sanguinetti M, Simren M, Gasbarrini A (2017) European consensus conference on faecal microbiota transplantation in clinical practice. Gut 66(4):569–580. https://doi.org/10.1136/gutjnl-2016-313017
49. Shouval R, Youngster I, Geva M, Eshel A, Danylesko I, Shimoni A, Beider K, Fein JA, Sharon I, Koren O, Nagler A (2018) Repeated courses of orally administered fecal microbiota transplantation for the treatment of steroid resistant and steroid dependent intestinal acute graft vs. host disease: a pilot study (NCT 03214289). Blood 132(1):2121–2121. https://doi.org/10.1182/blood-2018-99-110270

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.